O12 - Atomic Spectra

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Abstract

This experiment investigated the mercury (Hg) and sodium (Na) emission spectra through a reflective diffraction grating, focusing on the determination of the grating constant, the determination of wavelengths via angular measurements and the resolving power of the diffraction grating used. The grating constant was determined to be $g = (1971.57 \pm 86.14)$nm. While significant uncertainty arose primarily from high variation across measurements of the green emission line. The grating constant enabled the accurate reproduction of known spectral doublets from a sodium lamp, including the sodium D-lines at 589.07 ± 25.74 nm and 589.78 ± 25.77 nm, closely matching accepted literature values. However, the green doublet results suggested a yellowish shift not observed experimentally, the broad uncertainty range encompassed the expected green wavelengths. The resolving power of the grating was found to be $R = m(25360 \pm 1115)$, indicating a high resolution due to the small grating constant across an effective grating width of $W = 50$ mm. The results support the validity of the determined grating parameters, with majority of imprecision arising from grating constant variation. For future experimentation it is recommended for the Hg lamp to record more wavelengths in each mode, primarily to reduce the variation and increase the reliability of the grating constant.

1 Introduction

The emission spectra of elements provide critical insight into their atomic structure, with each element emitting light at characteristic wavelengths corresponding to electronic transitions. Spectral analysis of emission lines has widespread applications in fields such as chemical analysis, astronomy, and in medicine. In this experiment, the discrete emission spectra of mercury (Hg) and sodium (Na) were investigated using a reflective diffraction grating to disperse and measure the emitted spectral lines as well as investigate the diffraction grating used.

Hg vapor lamps are known for their distinct and well-separated emission lines, of interest in this investigation is the prominent green light at $\lambda \approx 546$ nm, which serves as a reliable calibration reference. By measuring the angular positions of this line in multiple diffraction orders, the grating constant of the reflective diffraction grating can be determined.

Na emission, on the other hand, is dominated by the well-known yellow D-lines, which result from the fine-structure transitions from the $3P_{1/2}$ and $3P_{3/2}$ excited states to the 3S ground state in sodium atoms. These lines occur at $\lambda = 589.0$ nm and $\lambda = 589.6$ nm and are widely used as a standard in spectroscopy due to their sharp definition and high intensity [2]. Additionally, sodium's emission spectrum includes several doublets in the visible range that was used in this investigation to assess the accuracy of determined wavelength.

The primary aim of this experiment was to analyze the Hg and Na emission spectra to determine the grating constant of the reflective diffraction grating, calculate the wavelengths of four sodium doublets using first-order spectra, and estimate the resolving power based on an effective grating width of $W = 50$ mm [3]. While the atomic spectra have important uses across the scientific community this investigation primarily focuses on the grating as a pivotal measurement technique and tool.

2 Hg Emission Spectra - Grating Constant
(Task 1)

2.1 Experimental Setup

Initially the Spectrometer, diffraction grating and high pressure Hg Lamp, was setup as indicated in, Fig. 1. to obtain via the precision spectrometer goniometer a zeroing or offset measurement to account for any deviations due to mechanical or user deviations by lining the internal scales of the ocular with the emission line of Hg.

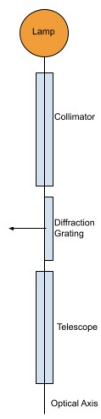

Figure 1: Initial Spectrometer setup for offset measurement

Next the observing telescope was rotated by approximating $\theta \approx 60°$ and the diffraction grating rotated by γ until the Hg emission line was visible again, indicating the angle of incidence α as outlined in Fig. 2. All relevant values were then measured using the internal scales of the observing telescope.
Next the observing telescope was further rotated and the angle, ϕ of the green ≈ 546 nm emission line for the 1st, 2nd and 3rd modes was measured, as indicated in Fig.3 Then the grating equation indicated below[3]:

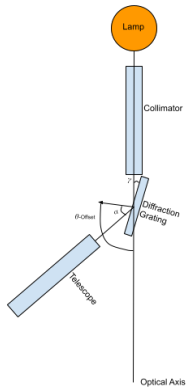

Figure 2: Angle of incidence measurement diagram

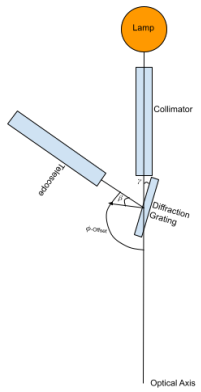

Figure 3: Angle of diffraction measurement diagram

$$
\begin{aligned}
g &= \frac{m\lambda}{\sin\beta - \sin\alpha} : \\
\alpha &= 90 - (\theta - \text{Offset}) - \gamma \\
\beta &= \gamma + (\phi - \text{Offset}) - 90
\end{aligned}
\tag{1}
$$

where α and β was determined from the measured angles $\theta, \phi, \text{Offset}, \text{and } \gamma$ respectively. The grating constant was determined for each mode and the average taken.

2.2 Data Analysis

Order	g **(nm)**
1	2017 ± 0.20
2	1850 ± 0.08
3	2046 ± 0.05
Avg±Std	1971.57 ± 86.14

Table 1: Grating Constant measurements for each order of 546.1 nm for Hg lamp

The results of the Hg emission spectra resulted in an average grating constant of $g = (1971.57 \pm 86.14)$nm. This value conveys a large grating constant (or coarse grating), and the significant uncertainty with relative error 4.9% is a result of a large spread of grating constants across the three modes of green ($\approx$ 546 nm) emission lines measured, which is predicted to be a result of inability to accurately determine the center of the emission line, or equivalently a consistent point of the emission line across modes.

3 Na Doublet Emission Spectra - Wavelengths

(Task II)

3.1 Experimental Setup

Initially the Mercury lamp was switched for a sodium (Na) lamp and then the same procedure as in Section: 2.1, for determination of the Offset and angle of incidence $\alpha \approx 60°$ was performed. Then the 4-Doublet emission lines for the first order diffraction, visually found to be blue, green, yellow and reddish/orange were located and their angle, ϕ, measured as previously outlined, for a total of 8 different measurements of β.
Through the rearrangement of Eq.(1). with $m = 1$, the wavelength λ of each emission line was determined with g set to be the determined value from the Hg Emission Spectra results.

3.2 Data Analysis

Based on the average grating constant determined in the previous task the wavelength of the doublets was determined to be:

Observed Colours	**Left λ (nm)**	**Right λ (nm)**
Blue	487.85 ± 21.31	488.28 ± 21.33
Greenish	560.29 ± 24.48	561.63 ± 24.54
Yellow	589.07 ± 25.74	589.78 ± 25.77
Reddish Orange	624.88 ± 27.30	625.72 ± 27.34

Table 2: Wavelength measurements for color pairs (Left and Right).

Viewing these wavelengths in the color spectrum, Fig.4., yields values that aligned with the colors observed during the experiment.

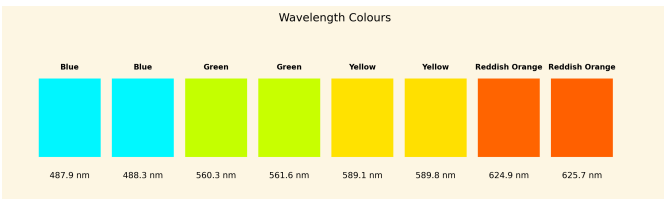

Figure 4: Plot of Wavelengths on the Colour Spectrum

With notable exception of the green wavelengths 560.3 nm and 561.6 nm which is expected to be higher than what was experimentally observed, resulting in a more yellowish tinge. However this discrepency does fall within the uncertainty of the particular wavelengths of 24.48 nm and 24.54 nm respectively. Notably the uncertainty of most of the wavelengths spans most of the range of its distinct color, e.g. the uncertainty in the green wavelengths covers most of the recognizable *green* color spectrum.

It is well known that the D-Lines of the Na emission spectra are located at 589.0 nm and 589.6 nm. The wavelengths in this experiment align closely with these values and when examining the color and the values uncertainty it is clear that experiment was able to locate these D-Lines given the accuracy of the tools available. The

large uncertainty is mostly a result of the propagation of the uncertainty in the grating constant $g = (1971.57 \pm 86.14)$nm. However factors such as the separation between doublets are a result of inability to find the center of emission line for each wavelength or equivalently measuring the emission lines from a consistent point across each wavelength. This factors explains the difference between the well-known 0.6 nm difference between the Na D-Lines and the 0.71 nm separation between yellow wavelengths determined from the experiment.

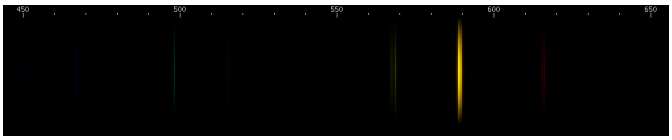

Figure 5: Emission Spectrum of Na

Source: [1]

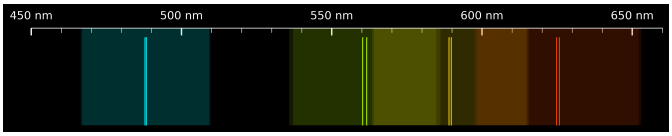

Figure 6: Experimental Emission Spectrum of Na

Fig.5. and Fig.6. provide a better visualization of the observed spectral lines of sodium emission against its emission spectrum. Each of the four observed doublets are located close to their expected positions with significant error that spans most, if not all of their respective color ranges. Of special significance are the D-lines of sodium, which closely align with their expected location.

4 Effective Grating Width - Resolving Power

Task III

Given an effective grating width $W = (50 \pm 5\%)$mm and the equations:

$$R = \frac{\lambda}{\Delta\lambda} = mN : N = \frac{W}{g} \tag{2}$$

The Resolving power was quickly determined to be:

$$R = m(25360 \pm 1115) : m \in \mathbb{Z}^+ \tag{3}$$

Indicating that is diffraction grating can distinguish a pair of wavelengths that vary by:

$$\Delta\lambda = \frac{\lambda}{R} \tag{4}$$

for the yellow wavelength 589.0 nm, this is $\approx$ 0.023 nm apart. A remarkably high resolution power.

5 Discussion

The results of this experiment were largely in-line with literature expectations, The determined grating constant was $g = (1971.57 \pm 86.14)$nm, expected to be on the larger side of grating constants. The high uncertainty was a result of the high variation in grating constants across the modes for the green $\approx$ 546 nm wavelengths, expected to be a result of a difficulty measuring a consistent point within the green emission line across modes. It is unknown as to whether the determined value for the grating is accurate, however based on the determined wavelengths of the 4 Doublets for the Na lamp, the grating constant determined was able to produce wavelengths that were in-line with experimental observations of blue, green,yellow and reddish orange doublets, (Table 2., Fig. 5,Fig. 6).

In particular the determined wavelengths for the sodium D-Lines, (589.07 ± 25.74)nm and (589.78 ± 25.77)nm were remarkably close to the well-known values of 589.0 and 589.6 nanometers, suggesting a degree of accuracy in the grating constant determined. The lack of precision in the determined wavelengths is primarily a result of the propagation of the high uncertainty, $Rel\ Err \approx 4.9\%$, of the grating constant. Addressing the green doublet, the determined values of (560.29 ± 24.48)nm and (561.63 ± 24.54)nm indicate a strong yellow tinge to the green doublet, however this yellow tinge was not identified during the experiment. Thus it is unknown if the green doublet is accurate, however due to the high uncertainty it is almost certain that the true doublet values lies within the uncertainty given the uncertainty spans a large portion of the green-light spectrum.

Finally the Resolving Power was determined to follow a trend of;

$$R = m(25360 \pm 1115) : m \in \mathbb{Z}^+$$

which is a very high resolution power, primarily due to the small grating constant over the effective width of 50mm. Although a significantly uncertainty due, once again to the large variation across grating constants determined.

6 Conclusion

The experiment successfully investigated the atomic emission spectra of mercury (Hg) and sodium (Na) using a reflective diffraction grating. Although the uncertainty (4.9%) was considerable, primarily due to variability in measurements, the results enabled accurate identification and reproduction of known sodium doublets, especially the prominent D-lines, which closely aligned with established values. Variations in grating constant measurements significantly impacted wavelength uncertainties, due to the inability to determine/measure a consistent measurement point across each wavelength.

The resolving power $R = m(25360 \pm 1115)$ indicated high spectral resolution capabilities, allowing differentiation of closely spaced wavelengths with a precision down to approximately 0.023 nm at the yellow sodium lines. The large uncertainty in the resolving power is a result of the high variation in the grating constant.

Overall, the experiment validated the method for spectral analysis using diffraction gratings, with results aligning well with theoretical expectations and known literature values. However the determined experimental values expressed a high uncertainty and questionable precision. For future experimentation it is recommended for the Hg

lamp to record more wavelengths in each mode, primarily to reduce the variation and increase the reliability of the grating constant.

References

[1] A. Kramida, G. Nave, and J. Reader. Sodium spectrum – nist atomic spectra database. https://atomic-spectra.net/spectrum.php?elem=Na&sw=430&lw=710, 2025. Accessed: 2025-06-03.

[2] National Institute of Standards and Technology. Strong Lines of Sodium (Na). https://physics.nist.gov/PhysRefData/Handbook/Tables/sodiumtable2.htm, 2025. Accessed: 2025-06-03.

[3] M. Ziese. *O12e Atomic Spectra*. Universität Leipzig, May 2025.